EARS Results — QA Report

File: **data/processed/dengue_ears.xlsx** · Built by ml/ears.py from dengue_cases_combined.xlsx · Report date: 18/10/2026

Summary

Calculation	**PASS** All 33,147 rows recomputed independently: 0 differences. Alarm labels, thresholds and notes are consistent.
2019 epidemic	**PASS** **Early, but only with DOH alongside.** The first national alarm was **26/05/2019** — 7 weeks before DOH's national alert (15/07/2019) and 10 weeks before the epidemic declaration (06/08/2019); all 17 regions alarmed before 15/07. But EARS alarms at the start of every season (mid-May to early June each year). What marks 2019 is the combination: all 13 national alarms in 2019 fell in DOH-epidemic weeks, against none of the 14 in 2020.
Agreement with DOH thresholds	**WARN** **Low week-by-week, high at onsets.** EARS and the DOH threshold agree on 62% of local weeks (κ = 0.19). That is expected: they answer different questions (sudden change vs. level compared with past years). At the start of DOH epidemic episodes, EARS fired before or at onset in 245 of 409 local episodes.
Watch-outs	**WARN** All 26 flagged one-week spikes raise a C1+C2+C3 alarm (artefacts, marked in flag_note). EARS stops alarming once an epidemic plateaus, and does not separate a large epidemic from an ordinary season start by alarm count alone.

1. What was run

Method	Baseline	Alarm when
C1	previous 7 weeks (t−7 … t−1)	(cases − mean) / SD > 3
C2	7 weeks with a 2-week gap (t−9 … t−3)	(cases − mean) / SD > 3
C3	C2 of this week and the 2 before	sum of max(0, C2 − 1) > 2

Settings agreed during cleaning: a week is usable only if data_quality = "ok" and it has no data_flag; a baseline needs 5 of its 7 weeks; flagged weeks still get their own result (with flag_note) but are left out of other weeks' baselines; SD is floored at 1 case and a week needs 5+ cases to alarm. ears_alarm reports which methods fired: C1, C2, C3, C1+C2, C1+C3, C2+C3, C1+C2+C3, none, or not computed. Details are on the METHOD sheet.

	NATIONAL (261 weeks)	LOCAL (32,886 place-weeks)
ears_alarm	none 144, not computed 49, C2+C3 38, C3 16, C1+C2+C3 12, C1 2	none 25,022, C3 2,908, C2+C3 1,939, not computed 1,444, C1+C2+C3 1,311, C1 197, C1+C2 39, C2 16, C1+C3 10

2. Integrity checks

Check	Result	
Row counts match the source workbook	261 national, 32,886 local	**PASS**
C1/C2/C3 recomputed with a separate implementation	33,147 rows, 0 differences	**PASS**
Every alarm meets its rule (stat > threshold, 5+ cases)	0 violations	**PASS**
ears_alarm label matches the C1/C2/C3 alarm columns	0 mismatches	**PASS**
Only allowed labels used	7 combinations + none + not computed	**PASS**
No result on missing / incomplete / backlog weeks	0 evaluated	**PASS**
Every flagged week carries a flag_note	279 of 279	**PASS**

3. The 2019 epidemic: EARS against the news

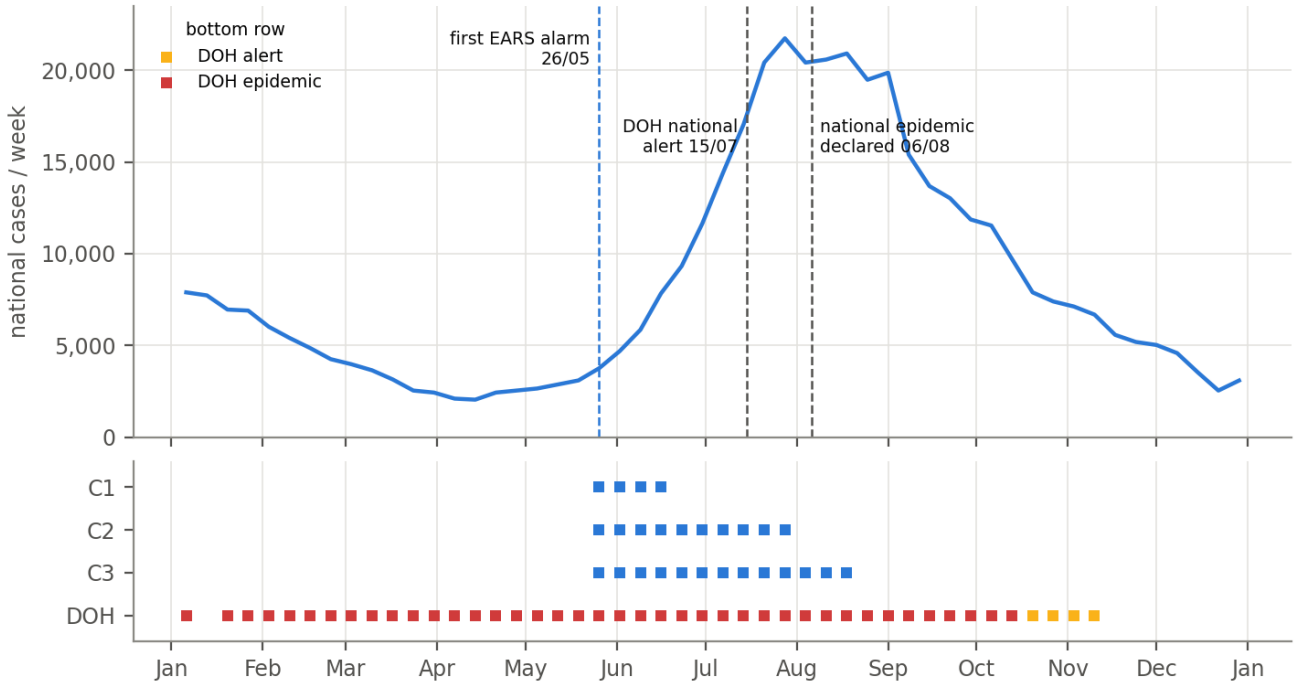


Figure 1. National weekly cases in 2019, with the weeks each EARS method alarmed and the DOH threshold status. Peak: 28/07/2019 (21,743 cases).

National EARS alarms began on **26/05/2019** (C1+C2+C3) as weekly cases passed 3,700, stayed on (C2+C3) through the steep rise of June–July, and switched off in late August once cases plateaued near 20,000 a week. The peak came on 28/07/2019, three weeks after the national alert.

Is 2019 different from an ordinary season?

Not to EARS on its own. Each year's season onset triggers alarms at about the same time, and the statistics were higher in 2016–2018, because those seasons rose from flatter baselines (a small SD makes the same rise look more extreme). EARS measures how sudden a rise is, not how big.

Year	First alarm	Alarm weeks	Max C1	Max C2	Max C3	Peak cases / week	DOH status during alarms
2016	29/05/2016	13	7.5	22.3	36.2	8,975	no DOH baseline
2017	04/06/2017	13	11.0	25.6	47.8	6,334	no DOH baseline
2018	13/05/2018	15	5.7	11.6	27.1	7,869	no DOH baseline
2019	26/05/2019	13	3.5	9.3	22.6	21,743	epidemic 13
2020	07/06/2020	14	4.1	6.4	13.0	6,354	no baseline 2, normal 12

Combined with the DOH threshold, 2019 stands out. Nationally, every 2019 alarm came while DOH was at "epidemic"; every 2020 alarm came while DOH was "normal". Locally (unflagged alarms), 86% of 2019 alarms coincided with DOH alert/epidemic against 27% in 2020. **Recommendation:** treat EARS as the timing signal (something just started) and the DOH threshold as the size signal (and it is unusually large); an EARS alarm during a DOH alert/epidemic week is the strongest early warning.

Timeline from news reports

Date	Event	Source
05/07/2019	Iloilo City declares a dengue outbreak	[1]
12/07/2019	Outbreaks declared in Capiz and Guimaras	[1]
15/07/2019	DOH raises a national dengue alert; MIMAROPA, Western Visayas, Central Visayas and Northern Mindanao past the epidemic threshold; 9 regions past the alert threshold. 106,630 cases 1 Jan – 29 Jun (+85% vs 2018)	[2] [3]
16/07/2019	Iloilo province placed under a state of calamity (6,806 cases, 23 deaths, Jan – 13 Jul)	[4]
Jul 2019	Zamboanga Sibugay state of calamity (3,005 cases Jan – 15 Jul vs 150 in 2018)	[5]
26/07/2019	Iloilo City placed under a state of calamity	[6]

Date	Event	Source
06/08/2019	DOH declares a national dengue epidemic: 146,062 cases and 622 deaths, 1 Jan – 20 Jul (+98% vs 2018)	[7] [8]
31/08/2019	271,480 cases and 1,107 deaths since 1 Jan	[9]
19/10/2019	371,717 cases and 1,407 deaths since 1 Jan (WHO situation report)	[10]
26/05/2019	**EARS (this analysis): first national alarm**	—

Places named in the news

Place	News event	Date	First EARS alarm 2019	Methods	Weeks ahead
Iloilo City	outbreak declared	05/07/2019	30/06/2019	C1+C2+C3	0
Capiz	outbreak declared	12/07/2019	19/05/2019	C1+C2+C3	7
Guimaras	outbreak declared	12/07/2019	02/06/2019	C2+C3	5
Iloilo	state of calamity	16/07/2019	26/05/2019	C1+C3	7
Zamboanga Sibugay	state of calamity	17/07/2019	03/03/2019	C2+C3	19
Iloilo City	state of calamity	26/07/2019	30/06/2019	C1+C2+C3	3

Weeks ahead = weeks from the first EARS alarm (from March 2019) to the news event. Zamboanga Sibugay already alarmed in early March, when its 2019 surge began.

4. Regions on 15 July 2019: DOH announcement vs. this analysis

On 15/07/2019 DOH named four regions past the epidemic threshold and nine past the alert threshold [2]. For comparison, local data was summed to regional totals and run through the same DOH-threshold and EARS calculations.

Region	DOH announced	DOH threshold (calculated)	First EARS alarm	Methods
REGION IV-B-MIMAROPA	epidemic	epidemic	16/06/2019	C1
REGION VI-WESTERN VISAYAS	epidemic	epidemic	02/06/2019	C1+C2+C3
REGION VII-CENTRAL VISAYAS	epidemic	alert	23/06/2019	C1+C2+C3
REGION X-NORTHERN MINDANAO	epidemic	epidemic	02/06/2019	C2+C3
BARMM	alert	alert	26/05/2019	C3
CAR	alert	alert	09/06/2019	C2+C3
REGION I-ILOCOS REGION	alert	alert	09/06/2019	C1
REGION II-CAGAYAN VALLEY	alert	alert	16/06/2019	C2+C3
REGION IV-A-CALABARZON	alert	epidemic	09/06/2019	C1+C2+C3
REGION IX-ZAMBOANGA PENINSULA	alert	epidemic	05/05/2019	C1+C2+C3
REGION V-BICOL REGION	alert	epidemic	16/06/2019	C1+C2+C3
REGION VIII-EASTERN VISAYAS	alert	epidemic	09/06/2019	C1
REGION XI-DAVAO REGION	alert	epidemic	09/06/2019	C1
CARAGA	not named	alert	30/06/2019	C2+C3
NATIONAL CAPITAL REGION	not named	normal	16/06/2019	C1
REGION III-CENTRAL LUZON	not named	normal	09/06/2019	C1+C2+C3
REGION XII-SOCCSKSARGEN	not named	alert	19/05/2019	C1

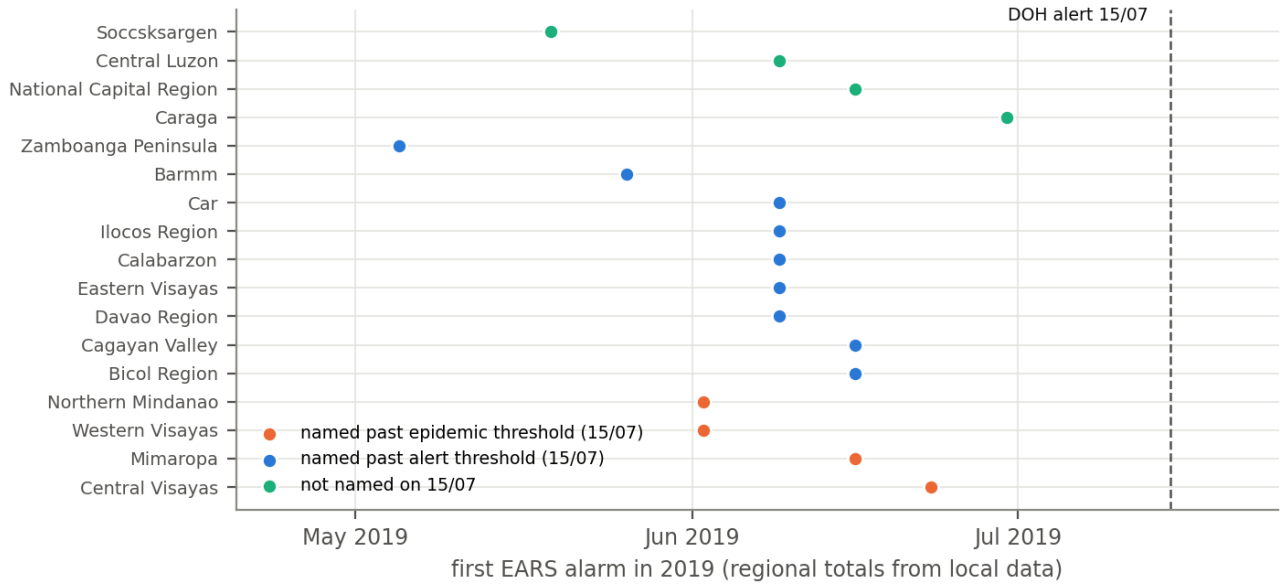


Figure 2. First EARS alarm in 2019 for each region, coloured by how DOH classified it on 15/07/2019.

Reading: every region alarmed before 15/07, including the four DOH did not name (NCR, Central Luzon, Caraga, Soccsksargen). EARS therefore gave early warning everywhere, but its first-alarm date does not rank severity: Zamboanga Peninsula and Soccsksargen alarmed first, while two of the four "epidemic" regions alarmed in mid/late June. The calculated DOH threshold matched the announced classification for 3 of 4 epidemic regions and 4 of 9 alert regions; the other 5 alert regions it rated "epidemic" — it runs hotter than the official system (§5).

5. Consistency with the DOH threshold columns

Week by week, 2019 onward, where both systems produced a result. "Elevated" = DOH alert or epidemic.

	Weeks	Agree	κ	EARS alarms when DOH elevated	DOH elevated when EARS alarms
National	85	48%	-0.01	29%	52%
Local	13,083	62%	0.19	28%	68%

Local ears_alarm	DOH normal	DOH alert	DOH epidemic
none	6,393	2,279	1,953
C1	24	26	27
C2	5	2	3
C3	370	306	406
C1+C2	8	10	16
C1+C3	1	2	1
C2+C3	214	207	342
C1+C2+C3	153	115	220

Why they disagree

• **Different questions.** DOH compares this week with the same week in the previous three years (is the level unusual for the season?). EARS compares it with the last 7–9 weeks (did it just jump?). During a long epidemic, DOH stays at "epidemic" while EARS falls silent once the rise levels off — nationally, DOH was "epidemic" every usable week from 06/01/2019, EARS only during the climb.

• **The calculated DOH threshold runs hot in 2019.** January–March 2019 averaged 5,056 national cases a week against 2,955 / 2,466 / 2,402 in 2016–2018, so the 3-year baseline put 2019 in "epidemic" from January — months before DOH announced anything. The official system uses 3–5 years of history and regional thresholds [2].

• **Onsets agree.** Of 409 local DOH epidemic episodes (2+ weeks), EARS alarmed within the 8 weeks before or during 281; in 245 of them the first alarm came before or at the episode's start (median 5 weeks ahead).

This is the comparison that matters for an early-warning system.

6. Watch-outs

WARN **Artefact alarms.** All 26 weeks flagged “one-week spike” raise C1+C2+C3. They are labelled in flag_note and should be excluded when counting true alarms.

WARN **Non-reporting weeks.** Of 253 “possible non-reporting” weeks, 97 were evaluated (all “none”) and 156 could not be computed because their baselines were also zeros. Results just after these runs rely on shortened baselines (see ears_note).

INFO **C3 on its own is the most frequent alarm** (2,908 local weeks). C3 accumulates three weeks of C2 signal, so it fires on sustained but modest rises; it is useful as an early hint but generates most of the alarm volume.

INFO **Alarm burden.** On average a place alarms 8–14 weeks a year (5.9 in 2020, when COVID-19 disrupted both dengue transmission and reporting). EARS is a screening tool: alarms prompt a look, not a declaration.

INFO **Small places** rely on the SD floor (2,311 notes) and the 5-case minimum (342 alarms suppressed); without them, single cases would trigger alarms.

Sources

[1] Manila Shimbun / NDRRMC — 89 dead from dengue in Western Visayas.
https://www.manila-shimbun.com/category/english/news245981.html

[2] ABS-CBN — Health department raises national dengue alert (15/07/2019).
https://news.abs-cbn.com/news/07/15/19/health-department-raises-national-dengue-alert

[3] Inquirer — DOH declares national dengue alert. https://newsinfo.inquirer.net/1141806/doh-declares-national-dengue-alert

[4] Inquirer — Iloilo placed under state of calamity amid rising dengue cases.
https://newsinfo.inquirer.net/1145599/iloilo-placed-under-state-of-calamity-amid-rising-dengue-cases

[5] Inquirer — Zamboanga Sibugay, Bicol bear brunt of dengue surge.
https://newsinfo.inquirer.net/1143014/zamboanga-sibugay-bicol-bear-brunt-of-dengue-surge/amp

[6] PTV — Iloilo City in state of calamity due to dengue. https://ptvnews.ph/?p=127418

[7] Inquirer — DOH declares national dengue epidemic (06/08/2019).
https://newsinfo.inquirer.net/1151103/doh-declares-national-dengue-epidemic

[8] Al Jazeera — Philippines declares national epidemic after 622 dengue deaths.
https://www.aljazeera.com/news/2019/08/philippines-declares-national-epidemic-622-dengue-deaths-190807010459368.html

[9] Gulf News — 1,107 deaths in Philippines from dengue epidemic. https://gulfnews.com/world/asia/philippines/1107-deaths-in-philippines-from-dengue-epidemic-113-higher-than-2018-1.1569153975766

[10] ReliefWeb / OCHA — Philippines: Situation Report 9, Dengue Outbreak (5 Nov 2019).
https://reliefweb.int/report/philippines/philippines-situation-report-9-dengue-outbreak-5-november-2019

News accessed 18/10/2026. Year-end totals vary by source and cut-off date, so only dated figures are quoted.